

Simple implementation of system to detect illegal dumping on campus using Raspberry Pi

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Introduction

We implemented a simple detection system using Raspberry Pi for the illegal dumping of household appliances that regularly occurred on the university campus (Fig. 1). Figure 2 shows the outline of our system. First, if a camera module attached to Raspberry Pi (for detail, see Fig. 3) detects persons, it captures and sends a photograph of them to the computation server. We assume the photograph is almost dark because captured at night, as shown in Fig. 4.

Therefore, we installed a deep learning algorithm for the nighttime visibility enhancement (for detail, see Fig. 5) on the computation server and applied it to the night photo. In the preliminary experiment shown in Fig. 6, we found that the enhanced nighttime photo contained a person and a chair.



Figure 1. Actual photo of illegal dumping on campus.

Materials



Figure 3. Contents of tupperware (Hardware).

Methods

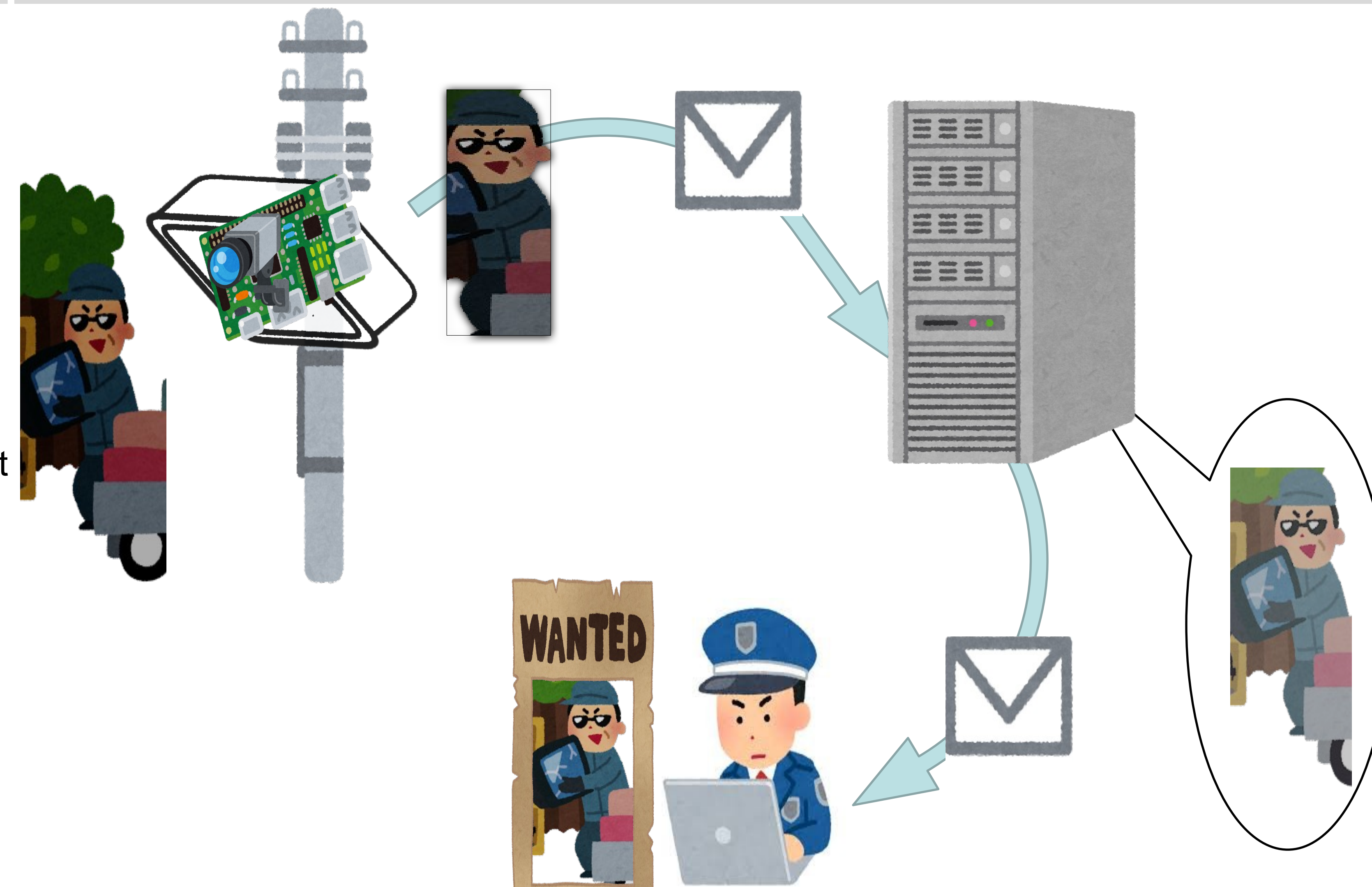


Figure 2. System overview.

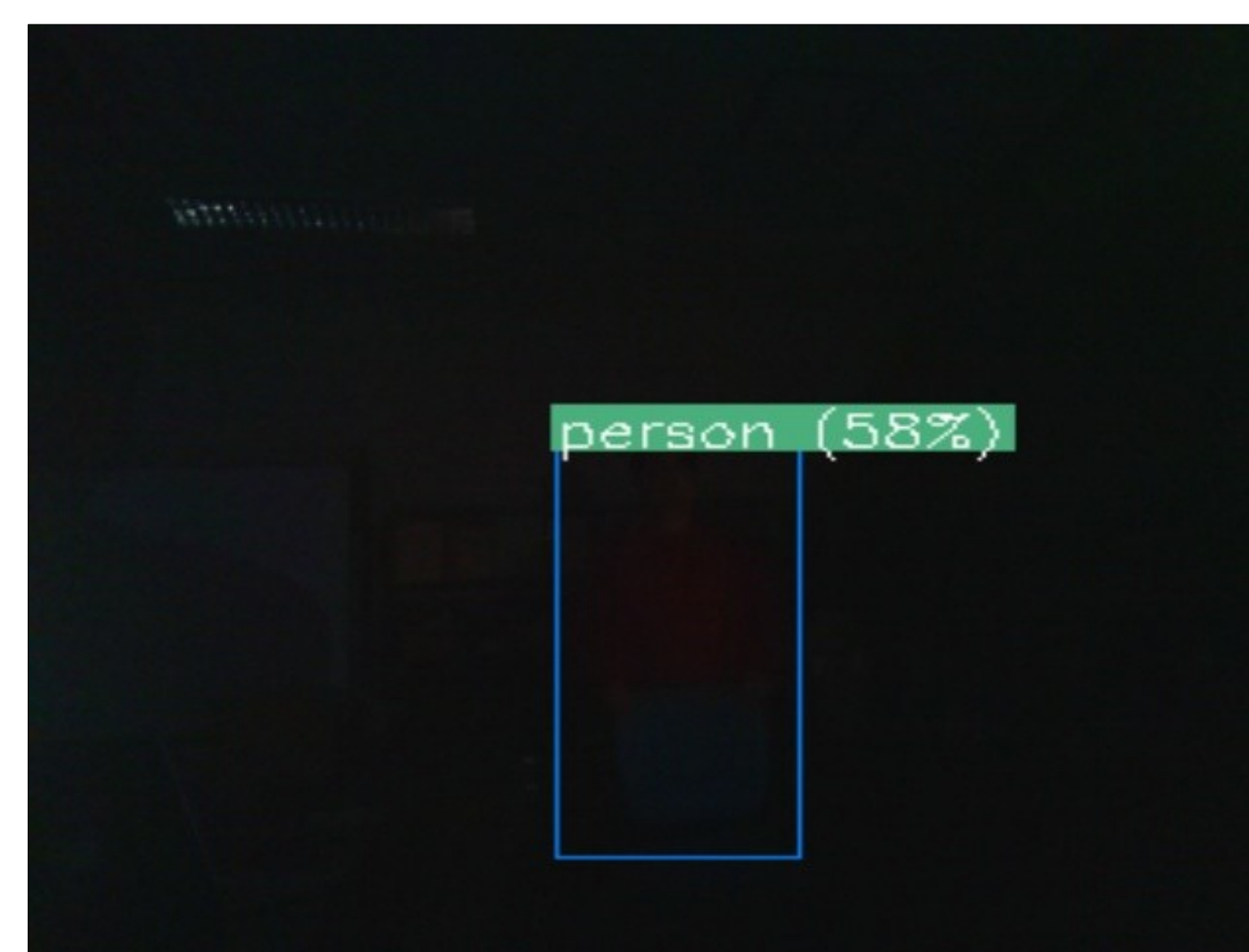


Figure 4. Human detection with TF-Lite, which could only detect a person

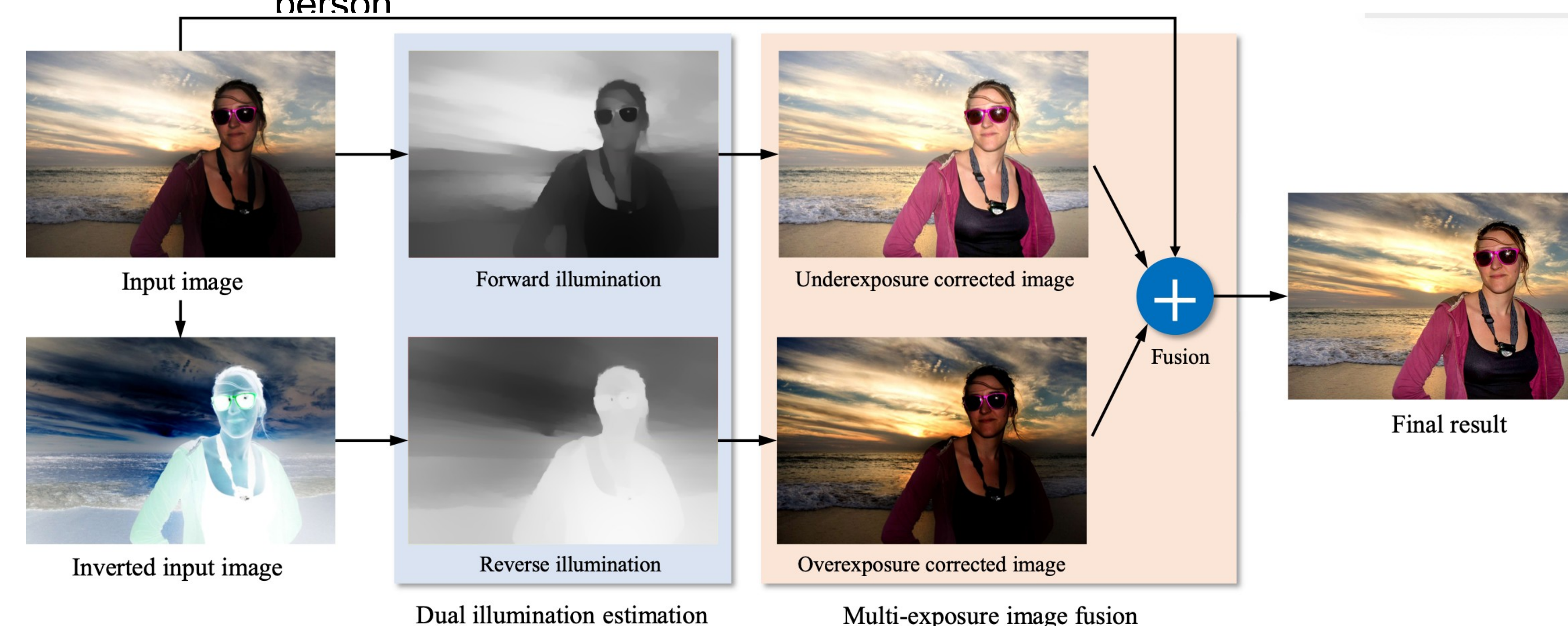


Figure 5. Deep learning algorithm for nighttime visibility enhancement (dual illumination estimation for robust exposure correction [3])

Results



Figure 6. Nighttime-visibility-enhanced image.

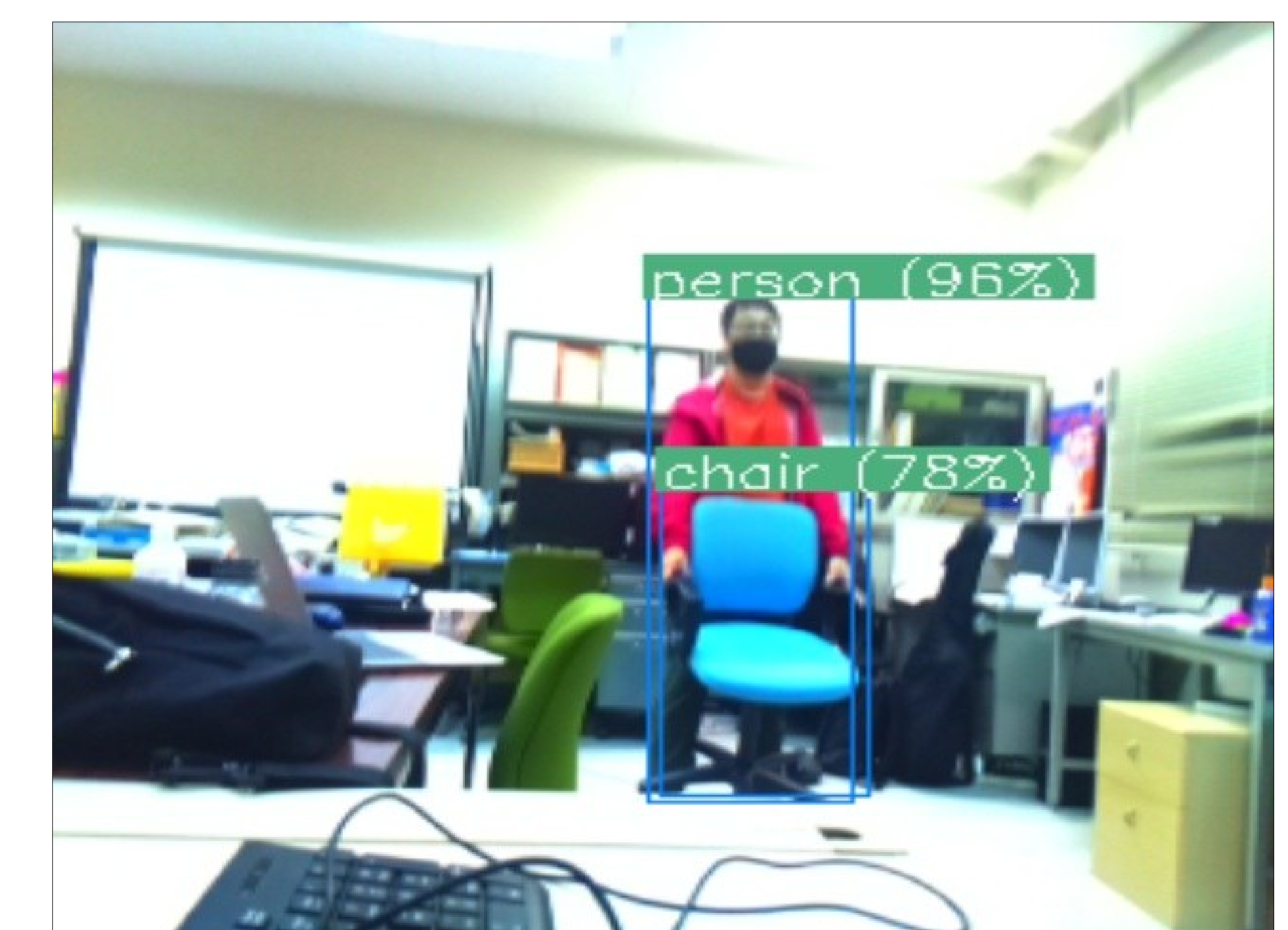


Figure 7. Ideal image.

References

1. https://sozorablog.com/camera_shooting/
2. https://www.tensorflow.org/lite/guide/python#install_tensorflow_lite_for_python
3. <https://arxiv.org/pdf/1910.13688.pdf>